# 光纤通道 API 方法

## Element Software

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# 光纤通道 API 方法

您可以使用光纤通道 API 方法添加，修改或删除存储集群的光纤通道节点成员。

- GetVolumeAccessGroupLunAssignments
- ListFibreChannelPortInfo
- ListFibreChannelSessions
- ListNodeFibreChannelPortInfo
- ModifyVolumeAccessGroupLunAssignments

## 了解更多信息

- "NetApp SolidFire 资源页面"
- "早期版本的 NetApp SolidFire 和 Element 产品的文档"

## GetVolumeAccessGroupLunAssignments

您可以使用 `GetVolumeAccessGroupLunAssignments` 方法检索有关指定卷访问组的 LUN 映射的详细信息。

### 参数

此方法具有以下输入参数：

| Name | Description | Type | 默认值 | Required |
|---|---|---|---|---|
| 卷访问组 ID | 用于返回信息的唯一卷访问组 ID 。 | 整型 | 无 | 是的。 |

### 返回值

此方法具有以下返回值：

| Name | Description | Type |
|---|---|---|
| volumeAccessGroupLunAssignments | 列出所有物理光纤通道端口或单个节点的端口。 | JSON 对象 |

### 请求示例

此方法的请求类似于以下示例：

```
{
   "method": "GetVolumeAccessGroupLunAssignments",
   "params": {
      "volumeAccessGroupID": 5
    },
  "id" : 1
   }
}
```

### 响应示例

此方法返回类似于以下示例的响应：

```
{
  "id" : 1,
  "result" : {
    "volumeAccessGroupLunAssignments" : {
       "volumeAccessGroupID" : 5,
       "lunAssignments" : [
          {"volumeID" : 5, "lun" : 0},
          {"volumeID" : 6, "lun" : 1},
          {"volumeID" : 7, "lun" : 2},
          {"volumeID" : 8, "lun" : 3}
       ],
       "deletedLunAssignments" : [
           {"volumeID" : 44, "lun" : 44}
       ]
    }
  }
}
```

### 自版本以来的新增功能

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## ListFibreChannelPortInfo

### 您可以使用 `ListFibreChannelPortInfo` 方法列出有关光纤通道端口的信息。

此 API 方法适用于单个节点；访问单个光纤通道节点需要用户 ID 和密码。但是，如果 force 参数设置为 true ，则可以在集群上使用此方法。如果在集群上使用，则会列出所有光纤通道接口。

## 参数

此方法具有以下输入参数：

| Name | Description | Type | 默认值 | Required |
|---|---|---|---|---|
| 【强制】 | 设置为 true 可在集群中的所有节点上运行。 | boolean | 无 | 否 |

## 返回值

此方法具有以下返回值：

| Name | Description | Type |
|---|---|---|
| FibreChannelPorts | 列出所有物理光纤通道端口或单个节点的端口。 | fibreChannelPort 数组 |

## 请求示例

此方法的请求类似于以下示例：

```
{
  "method": "ListFibreChannelPortInfo",
    "params": {},
    "id" : 1
}
```

## 响应示例

此方法返回类似于以下示例的响应：

```
{
  "id": 1,
  "result": {
    "fibreChannelPortInfo": {
      "5": {
        "result": {
          "fibreChannelPorts": [
            {
             "firmware": "7.04.00 (d0d5)",
             "hbaPort": 1,
             "model": "QLE2672",
             "nPortID": "0xc70084",
```

```
            "pciSlot": 3,
            "serial": "BFE1335E03500",
            "speed": "8 Gbit",
            "state": "Online",
            "switchWwn": "20:01:00:2a:6a:98:a3:41",
            "wwnn": "5f:47:ac:c8:3c:e4:95:00",
            "wwpn": "5f:47:ac:c0:3c:e4:95:0a"
          },
          {
            "firmware": "7.04.00 (d0d5)",
            "hbaPort": 2,
            "model": "QLE2672",
            "nPortID": "0x0600a4",
            "pciSlot": 3,
            "serial": "BFE1335E03500",
            "speed": "8 Gbit",
            "state": "Online",
            "switchWwn": "20:01:00:2a:6a:9c:71:01",
            "wwnn": "5f:47:ac:c8:3c:e4:95:00",
            "wwpn": "5f:47:ac:c0:3c:e4:95:0b"
          },
          {
            "firmware": "7.04.00 (d0d5)",
            "hbaPort": 1,
            "model": "QLE2672",
            "nPortID": "0xc70044",
            "pciSlot": 2,
            "serial": "BFE1335E04029",
            "speed": "8 Gbit",
            "state": "Online",
            "switchWwn": "20:01:00:2a:6a:98:a3:41",
            "wwnn": "5f:47:ac:c8:3c:e4:95:00",
            "wwpn": "5f:47:ac:c0:3c:e4:95:08"
        },
        {
          "firmware": "7.04.00 (d0d5)",
          "hbaPort": 2,
          "model": "QLE2672",
          "nPortID": "0x060044",
          "pciSlot": 2,
          "serial": "BFE1335E04029",
          "speed": "8 Gbit",
          "state": "Online",
          "switchWwn": "20:01:00:2a:6a:9c:71:01",
          "wwnn": "5f:47:ac:c8:3c:e4:95:00",
          "wwpn": "5f:47:ac:c0:3c:e4:95:09"
```

```
          }
        ]
      }
    },
    "6": {
      "result": {
          "fibreChannelPorts": [
        {
           "firmware": "7.04.00 (d0d5)",
           "hbaPort": 1,
           "model": "QLE2672",
           "nPortID": "0x060084",
           "pciSlot": 3,
           "serial": "BFE1335E04217",
           "speed": "8 Gbit",
           "state": "Online",
           "switchWwn": "20:01:00:2a:6a:9c:71:01",
           "wwnn": "5f:47:ac:c8:3c:e4:95:00",
           "wwpn": "5f:47:ac:c0:3c:e4:95:02"
        },
        {
           "firmware": "7.04.00 (d0d5)",
           "hbaPort": 2,
           "model": "QLE2672",
           "nPortID": "0xc700a4",
           "pciSlot": 3,
           "serial": "BFE1335E04217",
           "speed": "8 Gbit",
           "state": "Online",
           "switchWwn": "20:01:00:2a:6a:98:a3:41",
           "wwnn": "5f:47:ac:c8:3c:e4:95:00",
           "wwpn": "5f:47:ac:c0:3c:e4:95:03"
        },
        {
           "firmware": "7.04.00 (d0d5)",
           "hbaPort": 1,
           "model": "QLE2672",
           "nPortID": "0xc70064",
           "pciSlot": 2,
           "serial": "BFE1341E09515",
           "speed": "8 Gbit",
           "state": "Online",
           "switchWwn": "20:01:00:2a:6a:98:a3:41",
           "wwnn": "5f:47:ac:c8:3c:e4:95:00",
           "wwpn": "5f:47:ac:c0:3c:e4:95:00"
        },
```

```
          {
             "firmware": "7.04.00 (d0d5)",
             "hbaPort": 2,
             "model": "QLE2672",
             "nPortID": "0x060064",
             "pciSlot": 2,
             "serial": "BFE1341E09515",
             "speed": "8 Gbit",
             "state": "Online",
             "switchWwn": "20:01:00:2a:6a:9c:71:01",
             "wwnn": "5f:47:ac:c8:3c:e4:95:00",
             "wwpn": "5f:47:ac:c0:3c:e4:95:01"
          }
       ]
      }
     }
    }
   }
  }
```


### 自版本以来的新增功能

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## ListFibreChannelSessions

您可以使用 `ListFibreChannelSessions` 方法列出有关集群上光纤通道会话的信息。

### Parameters

此方法没有输入参数。

### 返回值

此方法具有以下返回值：

| Name | Description | Type |
|---|---|---|
| 会话 | 描述集群上活动光纤通道会话的对象列表。 | 会话 数组 |

### 请求示例

此方法的请求类似于以下示例：

```
{
   "method": "ListFibreChannelSessions",
   "params": {},
   "id" : 1
}
```

### 响应示例

此方法返回类似于以下示例的响应：

```
{
  "id" : 1,
  "result" : {
     "sessions" : [
    {
       "initiatorWWPN" : "21:00:00:0e:1e:14:af:40",
       "nodeID" : 5,
       "serviceID" : 21,
       "targetWWPN": "5f:47:ac:c0:00:00:00:10",
       "volumeAccessGroupID": 7
    },
    {
       "initiatorWWPN" : "21:00:00:0e:1e:14:af:40",
       "nodeID" : 1,
       "serviceID" : 22,
       "targetWWPN": "5f:47:ac:c0:00:00:00:11",
       "volumeAccessGroupID": 7
    }
    ]
  }
}
```

### 自版本以来的新增功能

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# ListNodeFibreChannelPortInfo

## 您可以使用 `ListNodeFibreChannelPortInfo` 方法列出有关节点上光纤通道端口的信息。

此 API 方法适用于单个节点；访问单个光纤通道节点需要用户 ID 和密码。如果在集群上使用，则会列出所有光纤通道接口。

## 参数

此方法没有输入参数。

## 返回值

此方法具有以下返回值：

| Name | Description | Type |
| --- | --- | --- |
| FibreChannelPorts | 列出所有物理光纤通道端口或单个节点的端口。 | fibreChannelPort 数组 |

## 请求示例

此方法的请求类似于以下示例：

```
{
   "method": "ListNodeFibreChannelPortInfo",
   "params": {
       "nodeID": 5,
       "force": true
   },
   "id" : 1
}
```

## 响应示例

此方法返回类似于以下示例的响应：

```
{
  "id": 1,
  "result": {
    "fibreChannelPorts": [
      {
        "firmware": "7.04.00 (d0d5)",
        "hbaPort": 1,
        "model": "QLE2672",
        "nPortID": "0xc7002c",
        "pciSlot": 3,
        "serial": "BFE1335E03500",
        "speed": "8 Gbit",
        "state": "Online",
        "switchWwn": "20:01:00:2a:6a:98:a3:41",
        "wwnn": "5f:47:ac:c8:35:54:02:00",
```

```
        "wwpn": "5f:47:ac:c0:35:54:02:02"
      },
      {
         "firmware": "7.04.00 (d0d5)",
         "hbaPort": 2,
         "model": "QLE2672",
         "nPortID": "0x06002d",
         "pciSlot": 3,
         "serial": "BFE1335E03500",
         "speed": "8 Gbit",
         "state": "Online",
         "switchWwn": "20:01:00:2a:6a:9c:71:01",
         "wwnn": "5f:47:ac:c8:35:54:02:00",
         "wwpn": "5f:47:ac:c0:35:54:02:03"
      },
      {
         "firmware": "7.04.00 (d0d5)",
         "hbaPort": 1,
         "model": "QLE2672",
         "nPortID": "0xc7002a",
         "pciSlot": 2,
         "serial": "BFE1335E04029",
         "speed": "8 Gbit",
         "state": "Online",
         "switchWwn": "20:01:00:2a:6a:98:a3:41",
         "wwnn": "5f:47:ac:c8:35:54:02:00",
         "wwpn": "5f:47:ac:c0:35:54:02:00"
     },
     {
         "firmware": "7.04.00 (d0d5)",
         "hbaPort": 2,
         "model": "QLE2672",
         "nPortID": "0x06002a",
         "pciSlot": 2,
         "serial": "BFE1335E04029",
         "speed": "8 Gbit",
         "state": "Online",
         "switchWwn": "20:01:00:2a:6a:9c:71:01",
         "wwnn": "5f:47:ac:c8:35:54:02:00",
         "wwpn": "5f:47:ac:c0:35:54:02:01"
      }
    ]
  }
}
```

## 自版本以来的新增功能

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# ModifyVolumeAccessGroupLunAssignments

## 您可以使用 `ModfyVolumeAccessGroupLunAssignments` 方法为特定卷定义自定义 LUN 分配。

此方法仅更改在卷访问组中 lunAssignments 参数上设置的 LUN 值。所有其他 LUN 分配均保持不变。

对于卷访问组中的卷， LUN 分配值必须是唯一的。您不能在卷访问组中定义重复的 LUN 值。但是，您可以在不同的卷访问组中再次使用相同的 LUN 值。

> 有效的 LUN 值为 0 到 16383 。如果传递的 LUN 值超出此范围，则系统会生成异常。如果存在异常，则不会修改任何指定的 LUN 分配。

小心

如果更改具有活动 I/O 的卷的 LUN 分配，则 I/O 可能会中断。在更改卷 LUN 分配之前，应更改服务器配置。

### Parameters

此方法具有以下输入参数：

| Name | Description | Type | 默认值 | Required |
|---|---|---|---|---|
| 卷访问组 ID | 要修改其 LUN 分配的唯一卷访问组 ID 。 | 整型 | 无 | 是的。 |
| lunAssignments | 具有新分配的 LUN 值的卷 ID 。 | 整型数组 | 无 | 是的。 |

## 返回值

此方法具有以下返回值：

| Name | Description | Type |
|---|---|---|
| volumeAccessGroupLunAssignments | 包含已修改卷访问组 LUN 分配的详细信息的对象。 | JSON 对象 |

## 请求示例

此方法的请求类似于以下示例：

```
{
   "method": "ModifyVolumeAccessGroupLunAssignments",
   "params": {
       "volumeAccessGroupID" : 218,
       "lunAssignments" : [
           {"volumeID" : 832, "lun" : 0},
           {"volumeID" : 834, "lun" : 1}
        ]
    },
    "id" : 1
}
```

## 响应示例

此方法返回类似于以下示例的响应：

```
{
  "id": 1,
  "result": {
    "volumeAccessGroupLunAssignments": {
      "deletedLunAssignments": [],
      "lunAssignments": [
        {
          "lun": 0,
          "volumeID": 832
        },
        {
          "lun": 1,
          "volumeID": 834
        }
      ],
      "volumeAccessGroupID": 218
    }
  }
}
```

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